

APPENDIX F

Complex Numbers

Operations with Complex Numbers • Complex Solutions of Quadratic Equations •
Polar Form of a Complex Number • Powers and Roots of Complex Numbers

Operations with Complex Numbers

Some equations have no real solutions. For instance, the quadratic equation

$$x^2 + 1 = 0 \quad \text{Equation with no real solution}$$

has no real solution because there is no real number x that can be squared to produce -1 . To overcome this deficiency, mathematicians created an expanded system of numbers using the **imaginary unit i** , defined as

$$i = \sqrt{-1} \quad \text{Imaginary unit}$$

where $i^2 = -1$. By adding real numbers to real multiples of this imaginary unit, you obtain the set of **complex numbers**. Each complex number can be written in the **standard form $a + bi$** .

Definition of a Complex Number

For real numbers a and b , the number

$$a + bi$$

is a **complex number**. If $b \neq 0$, $a + bi$ is called an **imaginary number**, and bi is called a **pure imaginary number**.

To add (or subtract) two complex numbers, you add (or subtract) the real and imaginary parts of the numbers separately.

Addition and Subtraction of Complex Numbers

If $a + bi$ and $c + di$ are two complex numbers written in standard form, their sum and difference are defined as follows.

$$\text{Sum: } (a + bi) + (c + di) = (a + c) + (b + d)i$$

$$\text{Difference: } (a + bi) - (c + di) = (a - c) + (b - d)i$$

The **additive identity** in the complex number system is zero (the same as in the real number system). Furthermore, the **additive inverse** of the complex number $a + bi$ is

$$-(a + bi) = -a - bi \quad \text{Additive inverse}$$

So, you have

$$(a + bi) + (-a - bi) = 0 + 0i = 0.$$

EXAMPLE 1 Adding and Subtracting Complex Numbers

- a.** $(3 - i) + (2 + 3i) = 3 - i + 2 + 3i$ Remove parentheses.
 $= 3 + 2 - i + 3i$ Group like terms.
 $= (3 + 2) + (-1 + 3)i$
 $= 5 + 2i$ Write in standard form.
- b.** $2i + (-4 - 2i) = 2i - 4 - 2i$ Remove parentheses.
 $= -4 + 2i - 2i$ Group like terms.
 $= -4$ Write in standard form.
- c.** $3 - (-2 + 3i) + (-5 + i) = 3 + 2 - 3i - 5 + i$
 $= 3 + 2 - 5 - 3i + i$
 $= 0 - 2i$
 $= -2i$

Notice in Example 1(b) that the sum of two complex numbers can be a real number.

Many of the properties of real numbers are valid for complex numbers as well. Here are some examples.

Associative Properties of Addition and Multiplication

Commutative Properties of Addition and Multiplication

Distributive Property of Multiplication over Addition

Notice below how these properties are used when two complex numbers are multiplied.

STUDY TIP Rather than trying to memorize the multiplication rule at the right, you can simply remember how the Distributive Property is used to multiply two complex numbers. The procedure is similar to multiplying two polynomials and combining like terms.

$$\begin{aligned} (a + bi)(c + di) &= a(c + di) + bi(c + di) && \text{Distributive Property} \\ &= ac + (ad)i + (bc)i + (bd)i^2 && \text{Distributive Property} \\ &= ac + (ad)i + (bc)i + (bd)(-1) && \text{Definition of } i \\ &= ac - bd + (ad)i + (bc)i && \text{Commutative Property} \\ &= (ac - bd) + (ad + bc)i && \text{Associative Property} \end{aligned}$$

EXAMPLE 2 Multiplying Complex Numbers

- a.** $(3 + 2i)(3 - 2i) = 9 - 6i + 6i - 4i^2$ Product of binomials
 $= 9 - 4(-1)$ $i^2 = -1$
 $= 9 + 4$ Simplify.
 $= 13$ Write in standard form.
- b.** $(3 + 2i)^2 = 9 + 6i + 6i + 4i^2$ Product of binomials
 $= 9 + 12i + 4(-1)$ $i^2 = -1$
 $= 9 - 4 + 12i$ Group like terms.
 $= 5 + 12i$ Write in standard form.

Notice in Example 2(a) that the product of two complex numbers can be a real number. This occurs with pairs of complex numbers of the form $a + bi$ and $a - bi$, called **complex conjugates**.

$$\begin{aligned}(a + bi)(a - bi) &= a^2 - abi + abi - b^2i^2 \\ &= a^2 - b^2(-1) \\ &= a^2 + b^2\end{aligned}$$

To write the quotient of $a + bi$ and $c + di$ in standard form, where c and d are not both zero, multiply the numerator and denominator by the complex conjugate of the denominator to obtain

$$\begin{aligned}\frac{a + bi}{c + di} &= \frac{a + bi}{c + di} \left(\frac{c - di}{c - di} \right) \\ &= \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2}.\end{aligned}$$

EXAMPLE 3 Writing a Quotient of Complex Numbers in Standard Form

$$\begin{aligned}\frac{2 + 3i}{4 - 2i} &= \frac{2 + 3i}{4 - 2i} \left(\frac{4 + 2i}{4 + 2i} \right) && \text{Multiply numerator and denominator by complex conjugate of denominator.} \\ &= \frac{8 + 4i + 12i + 6i^2}{16 - 4i^2} && \text{Expand.} \\ &= \frac{8 - 6 + 16i}{16 + 4} && i^2 = -1 \\ &= \frac{2 + 16i}{20} && \text{Simplify.} \\ &= \frac{1}{10} + \frac{4}{5}i && \text{Write in standard form.}\end{aligned}$$

Complex Solutions of Quadratic Equations

When using the Quadratic Formula to solve a quadratic equation, you often obtain a result such as $\sqrt{-3}$, which you know is not a real number. By factoring out $i = \sqrt{-1}$, you can write this number in standard form.

$$\sqrt{-3} = \sqrt{3(-1)} = \sqrt{3}\sqrt{-1} = \sqrt{3}i$$

The number $\sqrt{3}i$ is called the *principal square root* of -3 .

STUDY TIP The definition of principal square root uses the rule

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

for $a > 0$ and $b < 0$. This rule is not valid if *both* a and b are negative.

For example,

$$\begin{aligned}\sqrt{-5}\sqrt{-5} &= \sqrt{5(-1)}\sqrt{5(-1)} \\ &= \sqrt{5}i\sqrt{5}i \\ &= \sqrt{25}i^2 \\ &= 5i^2 = -5\end{aligned}$$

whereas

$$\sqrt{(-5)(-5)} = \sqrt{25} = 5.$$

To avoid problems with multiplying square roots of negative numbers, be sure to convert to standard form *before* multiplying.

Principal Square Root of a Negative Number

If a is a positive number, the **principal square root** of the negative number $-a$ is defined as

$$\sqrt{-a} = \sqrt{a}i.$$

EXAMPLE 4 Writing Complex Numbers in Standard Form

$$\begin{aligned}\text{a. } \sqrt{-3}\sqrt{-12} &= \sqrt{3}i\sqrt{12}i \\ &= \sqrt{36}i^2 \\ &= 6(-1) \\ &= -6\end{aligned}$$

$$\begin{aligned}\text{b. } \sqrt{-48} - \sqrt{-27} &= \sqrt{48}i - \sqrt{27}i \\ &= 4\sqrt{3}i - 3\sqrt{3}i \\ &= \sqrt{3}i\end{aligned}$$

$$\begin{aligned}\text{c. } (-1 + \sqrt{-3})^2 &= (-1 + \sqrt{3}i)^2 \\ &= (-1)^2 - 2\sqrt{3}i + (\sqrt{3})^2(i^2) \\ &= 1 - 2\sqrt{3}i + 3(-1) \\ &= -2 - 2\sqrt{3}i\end{aligned}$$

EXAMPLE 5 Complex Solutions of a Quadratic Equation

Solve $3x^2 - 2x + 5 = 0$.

Solution

$$\begin{aligned}x &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)(5)}}{2(3)} \\ &= \frac{2 \pm \sqrt{-56}}{6} \\ &= \frac{2 \pm 2\sqrt{14}i}{6} \\ &= \frac{1}{3} \pm \frac{\sqrt{14}}{3}i\end{aligned}$$

Quadratic Formula

Simplify.

Write in i -form.

Write in standard form.

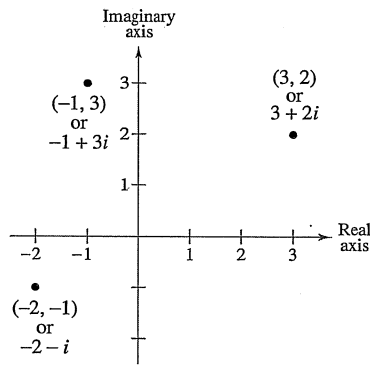


Figure F.1

Polar Form of a Complex Number

Just as real numbers can be represented by points on the real number line, you can represent a complex number

$$z = a + bi$$

as the point (a, b) in a coordinate plane (the **complex plane**). The horizontal axis is called the **real axis** and the vertical axis is called the **imaginary axis**, as shown in Figure F.1.

The **absolute value** of a complex number $a + bi$ is defined as the distance between the origin $(0, 0)$ and the point (a, b) .

The Absolute Value of a Complex Number

The **absolute value** of the complex number $z = a + bi$ is given by

$$|a + bi| = \sqrt{a^2 + b^2}.$$

If the complex number $a + bi$ is a real number (that is, if $b = 0$), then this definition agrees with that given for the absolute value of a real number.

$$|a + 0i| = \sqrt{a^2 + 0^2} = |a|.$$

To work effectively with *powers* and *roots* of complex numbers, it is helpful to write complex numbers in **polar form**. In Figure F.2, consider the nonzero complex number $a + bi$. By letting θ be the angle from the positive real axis (measured counterclockwise) to the line segment connecting the origin and the point (a, b) , you can write

$$a = r \cos \theta \quad \text{and} \quad b = r \sin \theta$$

where $r = \sqrt{a^2 + b^2}$. Consequently, you have

$$a + bi = (r \cos \theta) + (r \sin \theta)i$$

from which you can obtain the **polar form of a complex number**.

Polar Form of a Complex Number

The **polar form** of the complex number $z = a + bi$ is given by

$$z = r(\cos \theta + i \sin \theta) = r e^{i\theta} \quad (*) \quad \text{Euler's Formula}$$

where $a = r \cos \theta$, $b = r \sin \theta$, $r = \sqrt{a^2 + b^2}$, and $\tan \theta = b/a$. The number r is the **modulus** of z , and θ is called an **argument** of z .

NOTE The polar form of a complex number is also called the **trigonometric form**. Because there are infinitely many choices for θ , the polar form of a complex number is not unique. Normally, θ is restricted to the interval $0 \leq \theta < 2\pi$, although on occasion it is convenient to use $\theta < 0$.

(*) Please see the other handout for a description of Euler's formula.

EXAMPLE 6 Writing a Complex Number in Polar Form

Write the complex number $z = -2 - 2\sqrt{3}i$ in polar form.

Solution The absolute value of z is

$$r = |-2 - 2\sqrt{3}i| = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{16} = 4$$

and the angle θ is given by

$$\begin{aligned}\tan \theta &= \frac{b}{a} \\ &= \frac{-2\sqrt{3}}{-2} \\ &= \sqrt{3}.\end{aligned}$$

Because $\tan(\pi/3) = \sqrt{3}$ and because $z = -2 - 2\sqrt{3}i$ lies in Quadrant III, choose θ to be $\theta = \pi + \pi/3 = 4\pi/3$. So, the polar form is

$$\begin{aligned}z &= r(\cos \theta + i \sin \theta) \\ &= 4\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right).\end{aligned}$$

See Figure F.3.

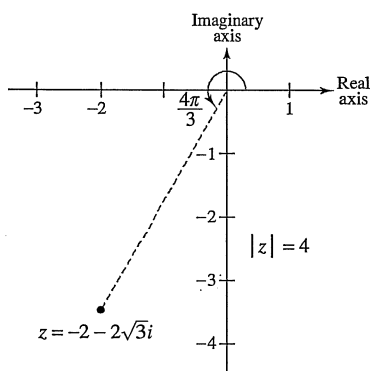


Figure F.3

The polar form adapts nicely to multiplication and division of complex numbers. Suppose you are given two complex numbers

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1) \quad \text{and} \quad z_2 = r_2(\cos \theta_2 + i \sin \theta_2).$$

The product of z_1 and z_2 is

$$\begin{aligned}z_1 z_2 &= r_1 r_2 (\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)].\end{aligned}$$

Using the sum and difference formulas for cosine and sine, you can rewrite this equation as

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)].$$

This establishes the first part of the following rule. Try to establish the second part on your own.

Product and Quotient of Two Complex Numbers

Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$ be complex numbers.

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)] \quad \text{Product}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)], \quad z_2 \neq 0 \quad \text{Quotient}$$

Note that this rule says that to multiply two complex numbers you multiply moduli and add arguments, whereas to divide two complex numbers you divide moduli and subtract arguments.

EXAMPLE 7 Multiplying Complex Numbers in Polar Form

Find the product $z_1 z_2$ of the complex numbers.

$$z_1 = 2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right), \quad z_2 = 8\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$$

Solution

$$\begin{aligned} z_1 z_2 &= 2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right) \cdot 8\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right) \\ &= 16\left[\cos\left(\frac{2\pi}{3} + \frac{11\pi}{6}\right) + i \sin\left(\frac{2\pi}{3} + \frac{11\pi}{6}\right)\right] \\ &= 16\left[\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2}\right] \\ &= 16\left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right] \\ &= 16[0 + i(1)] \\ &= 16i \end{aligned}$$

Check this result by first converting to the standard forms $z_1 = -1 + \sqrt{3}i$ and $z_2 = 4\sqrt{3} - 4i$ and then multiplying algebraically.

EXAMPLE 8 Dividing Complex Numbers in Polar Form

Find the quotient z_1/z_2 of the complex numbers.

$$z_1 = 24(\cos 300^\circ + i \sin 300^\circ), \quad z_2 = 8(\cos 75^\circ + i \sin 75^\circ)$$

Solution

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{24(\cos 300^\circ + i \sin 300^\circ)}{8(\cos 75^\circ + i \sin 75^\circ)} \\ &= \frac{24}{8} [\cos(300^\circ - 75^\circ) + i \sin(300^\circ - 75^\circ)] \\ &= 3[\cos 225^\circ + i \sin 225^\circ] \\ &= 3\left[\left(-\frac{\sqrt{2}}{2}\right) + i\left(-\frac{\sqrt{2}}{2}\right)\right] \\ &= -\frac{3\sqrt{2}}{2} - \frac{3\sqrt{2}}{2}i \end{aligned}$$

Powers and Roots of Complex Numbers

To raise a complex number to a power, consider repeated use of the multiplication rule.

$$z = r(\cos \theta + i \sin \theta)$$

$$z^2 = r^2(\cos 2\theta + i \sin 2\theta)$$

$$z^3 = r^3(\cos 3\theta + i \sin 3\theta)$$

$$\vdots$$

This pattern leads to the following important theorem, which is named after the French mathematician Abraham DeMoivre (1667–1754).

THEOREM F.1 DeMoivre's Theorem

If $z = r(\cos \theta + i \sin \theta)$ is a complex number and n is a positive integer, then

$$z^n = [r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta) = r^n e^{in\theta}$$

⊗ Euler's Formula

EXAMPLE 9 Finding Powers of a Complex Number

Use DeMoivre's Theorem to find $(-1 + \sqrt{3}i)^{12}$.

Solution First convert to polar form.

$$-1 + \sqrt{3}i = 2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$$

Then, by DeMoivre's Theorem, you have

$$\begin{aligned} (-1 + \sqrt{3}i)^{12} &= \left[2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)\right]^{12} \\ &= 2^{12} \left[\cos \left(12 \cdot \frac{2\pi}{3}\right) + i \sin \left(12 \cdot \frac{2\pi}{3}\right) \right] \\ &= 4096(\cos 8\pi + i \sin 8\pi) \\ &= 4096. \end{aligned}$$

NOTE Notice in Example 9 that the answer is a real number.

Recall that a consequence of the Fundamental Theorem of Algebra is that a polynomial equation of degree n has n solutions in the complex number system. Each solution is an n th root of the equation. The n th root of a complex number is defined as follows.

Definition of n th Root of a Complex Number

The complex number $u = a + bi$ is an n th root of the complex number z if

$$z = u^n = (a + bi)^n.$$

STUDY TIP The n th roots of a complex number are useful for solving some polynomial equations. For instance, you can use DeMoivre's Theorem to solve the polynomial equation

$$x^4 + 16 = 0$$

by writing -16 as $16(\cos \pi + i \sin \pi)$.

To find a formula for an n th root of a complex number, let u be an n th root of z , where

$$u = s(\cos \beta + i \sin \beta) \quad \text{and} \quad z = r(\cos \theta + i \sin \theta).$$

By DeMoivre's Theorem and the fact that $u^n = z$, you have

$$s^n(\cos n\beta + i \sin n\beta) = r(\cos \theta + i \sin \theta).$$

Taking the absolute value of each side of this equation, it follows that $s^n = r$. Substituting back into the previous equation and dividing by r , you get

$$\cos n\beta + i \sin n\beta = \cos \theta + i \sin \theta.$$

So, it follows that

$$\cos n\beta = \cos \theta \quad \text{and} \quad \sin n\beta = \sin \theta.$$

Because both sine and cosine have a period of 2π , these last two equations have solutions if and only if the angles differ by a multiple of 2π . Consequently, there must exist an integer k such that

$$n\beta = \theta + 2\pi k$$

$$\beta = \frac{\theta + 2\pi k}{n}.$$

By substituting this value for β into the polar form of u , you get the following result.

THEOREM F.2 n th Roots of a Complex Number

For a positive integer n , the complex number $z = r(\cos \theta + i \sin \theta)$ has exactly n distinct n th roots given by

$$\sqrt[n]{r} \left(\cos \frac{\theta + 2\pi k}{n} + i \sin \frac{\theta + 2\pi k}{n} \right) = r^{\frac{1}{n}} e^{i \left(\frac{\theta + 2\pi k}{n} \right)}$$

ⓧ Euler's Formula

where $k = 0, 1, 2, \dots, n - 1$.

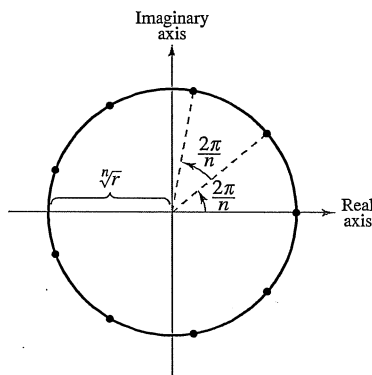


Figure F.4

When k exceeds $n - 1$, the roots begin to repeat. For instance, if $k = n$, the angle

$$\frac{\theta + 2\pi n}{n} = \frac{\theta}{n} + 2\pi$$

is coterminal with θ/n , which is also obtained when $k = 0$.

This formula for the n th roots of a complex number z has a nice geometric interpretation, as shown in Figure F.4. Note that because the n th roots of z all have the same magnitude $\sqrt[n]{r}$, they all lie on a circle of radius $\sqrt[n]{r}$ with center at the origin. Furthermore, because successive n th roots have arguments that differ by $2\pi/n$, the n roots are equally spaced along the circle.

EXAMPLE 10 Finding the n th Roots of a Complex NumberFind the three cube roots of $z = -2 + 2i$.**Solution**Because z lies in Quadrant II, the polar form for z is

$$z = -2 + 2i = \sqrt{8}(\cos 135^\circ + i \sin 135^\circ).$$

By the formula for n th roots, the cube roots have the form

$$\sqrt[3]{8} \left(\cos \frac{135^\circ + 360^\circ k}{3} + i \sin \frac{135^\circ + 360^\circ k}{3} \right).$$

Finally, for $k = 0, 1$, and 2 , you obtain the roots

$$\sqrt{2}(\cos 45^\circ + i \sin 45^\circ) = 1 + i$$

$$\sqrt{2}(\cos 165^\circ + i \sin 165^\circ) \approx -1.3660 + 0.3660i$$

$$\sqrt{2}(\cos 285^\circ + i \sin 285^\circ) \approx 0.3660 - 1.3660i.$$

EXERCISES FOR APPENDIX F

In Exercises 1–24, perform the operation and write the result in standard form.

1. $(5 + i) + (6 - 2i)$
 2. $(13 - 2i) + (-5 + 6i)$
 3. $(8 - i) - (4 - i)$
 4. $(3 + 2i) - (6 + 13i)$
 5. $(-2 + \sqrt{-8}) + (5 - \sqrt{-50})$
 6. $(8 + \sqrt{-18}) - (4 + 3\sqrt{2}i)$
 7. $13i - (14 - 7i)$
 8. $22 + (-5 + 8i) + 10i$
 9. $-\left(\frac{3}{2} + \frac{5}{2}i\right) + \left(\frac{5}{3} + \frac{11}{3}i\right)$
 10. $(1.6 + 3.2i) + (-5.8 + 4.3i)$
 11. $\sqrt{-6} \cdot \sqrt{-2}$
 12. $\sqrt{-5} \cdot \sqrt{-10}$
 13. $(\sqrt{-10})^2$
 14. $(\sqrt{-75})^2$
 15. $(1 + i)(3 - 2i)$
 16. $(6 - 2i)(2 - 3i)$
 17. $6i(5 - 2i)$
 18. $-8i(9 + 4i)$
 19. $(\sqrt{14} + \sqrt{10}i)(\sqrt{14} - \sqrt{10}i)$
 20. $(3 + \sqrt{-5})(7 - \sqrt{-10})$
 21. $(4 + 5i)^2$
 22. $(2 - 3i)^2$
 23. $(2 + 3i)^2 + (2 - 3i)^2$
 24. $(1 - 2i)^2 - (1 + 2i)^2$

In Exercises 25–32, write the complex conjugate of the complex number. Then multiply the number by its complex conjugate.

25. $5 + 3i$
 26. $9 - 12i$
 27. $-2 - \sqrt{5}i$
 28. $-4 + \sqrt{2}i$
 29. $20i$
 30. $\sqrt{-15}$
 31. $\sqrt{8}$
 32. $1 + \sqrt{8}$

See defn
on pg F3

In Exercises 33–42, write the quotient in standard form.

33. $\frac{6}{i}$
 34. $-\frac{10}{2i}$
 35. $\frac{4}{4 - 5i}$
 36. $\frac{3}{1 - i}$
 37. $\frac{2 + i}{2 - i}$
 38. $\frac{8 - 7i}{1 - 2i}$
 39. $\frac{6 - 7i}{i}$
 40. $\frac{8 + 20i}{2i}$
 41. $\frac{1}{(4 - 5i)^2}$
 42. $\frac{(2 - 3i)(5i)}{2 + 3i}$

See
Ex 3

In Exercises 43–46, perform the operation and write the result in standard form.

43. $\frac{2}{1 + i} - \frac{3}{1 - i}$
 44. $\frac{2i}{2 + i} + \frac{5}{2 - i}$
 45. $\frac{i}{3 - 2i} + \frac{2i}{3 + 8i}$
 46. $\frac{1 + i}{i} - \frac{3}{4 - i}$

See
Ex 1
Ex 2
Ex 4

In Exercises 47–54, use the Quadratic Formula to solve the quadratic equation.

47. $x^2 - 2x + 2 = 0$
 48. $x^2 + 6x + 10 = 0$
 49. $4x^2 + 16x + 17 = 0$
 50. $9x^2 - 6x + 37 = 0$
 51. $4x^2 + 16x + 15 = 0$
 52. $9x^2 - 6x - 35 = 0$
 53. $16t^2 - 4t + 3 = 0$
 54. $5s^2 + 6s + 3 = 0$

In Exercises 55–62, simplify the complex number and write it in standard form.

55. $-6i^3 + i^2$
 56. $4i^2 - 2i^3$
 57. $-5i^5$
 58. $(-i)^3$
 59. $(\sqrt{-75})^3$
 60. $(\sqrt{-2})^6$
 61. $\frac{1}{i^3}$
 62. $\frac{1}{(2i)^3}$

Use: $i^2 = -1$
 $i^3 = -i$
 $i^4 = 1$

In Exercises 63–68, plot the complex number and find its absolute value.

63. $-5i$
 64. -5
 65. $-4 + 4i$
 66. $5 - 12i$
 67. $6 - 7i$
 68. $-8 + 3i$

In Exercises 69–76, represent the complex number graphically, and find the polar form of the number.

69. $3 - 3i$
 70. $2 + 2i$
 71. $\sqrt{3} + i$
 72. $-1 + \sqrt{3}i$
 73. $-2(1 + \sqrt{3}i)$
 74. $\frac{5}{2}(\sqrt{3} - i)$
 75. $6i$
 76. 4

In Exercises 77–82, represent the complex number graphically, and find the standard form of the number.

77. $2(\cos 150^\circ + i \sin 150^\circ)$
 78. $5(\cos 135^\circ + i \sin 135^\circ)$
 79. $\frac{3}{2}(\cos 300^\circ + i \sin 300^\circ)$
 80. $\frac{3}{4}(\cos 315^\circ + i \sin 315^\circ)$
 81. $3.75\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$
 82. $8\left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12}\right)$

In Exercises 83–86, perform the operation and leave the result in polar form.

83. $\left[3\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)\right]\left[4\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)\right]$
 84. $\left[\frac{3}{2}\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)\right]\left[6\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)\right]$
 85. $\left[\frac{5}{3}(\cos 140^\circ + i \sin 140^\circ)\right]\left[\frac{2}{3}(\cos 60^\circ + i \sin 60^\circ)\right]$
 86. $\frac{\cos(5\pi/3) + i \sin(5\pi/3)}{\cos \pi + i \sin \pi}$

In Exercises 87–94, use DeMoivre's Theorem to find the indicated power of the complex number. Write the result in standard form.

87. $(1 + i)^5$
 88. $(2 + 2i)^6$
 89. $(-1 + i)^{10}$
 90. $(1 - i)^{12}$
 91. $2(\sqrt{3} + i)^7$
 92. $4(1 - \sqrt{3}i)^3$
 93. $\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)^{10}$
 94. $\left[2\left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)\right]^8$

In Exercises 95–100, (a) use Theorem F.2 on page F9 to find the indicated roots of the complex number, (b) represent each of the roots graphically, and (c) write each of the roots in standard form.

95. Square roots of $5(\cos 120^\circ + i \sin 120^\circ)$
 96. Square roots of $16(\cos 60^\circ + i \sin 60^\circ)$
 97. Fourth roots of $16\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right)$
 98. Fifth roots of $32\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$
 99. Cube roots of $-\frac{125}{2}(1 + \sqrt{3}i)$
 100. Cube roots of $-4\sqrt{2}(1 - i)$

In Exercises 101–108, use Theorem F.2 on page F9 to find all the solutions of the equation and represent the solutions graphically.

101. $x^4 - i = 0$
 102. $x^3 + 1 = 0$
 103. $x^5 + 243 = 0$
 104. $x^4 - 81 = 0$
 105. $x^3 + 64i = 0$
 106. $x^6 - 64i = 0$
 107. $x^3 - (1 - i) = 0$
 108. $x^4 + (1 + i) = 0$

Solve for x and use Thm F.2

◇ Real and Imaginary parts of a complex number:

Let $z = a + bi$. Then the **real and imaginary parts** of z are denoted as

$$\operatorname{Re}(z) = a \quad \text{and} \quad \operatorname{Im}(z) = b$$

◇ **Euler's Formula:** $e^{i\theta} = \cos \theta + i \sin \theta$

◇ These ideas lead to the following very important identities for $\cos \theta$ and $\sin \theta$.

$$\cos \theta = \operatorname{Re}(e^{i\theta}) = \frac{e^{i\theta} + e^{-i\theta}}{2} \quad \text{and} \quad \sin \theta = \operatorname{Im}(e^{i\theta}) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

◇ Euler's formula gives another way to view the polar form of a complex number.

Recall that $z = a + bi$ can be written in the following **polar form**

$$z = r(\cos \theta + i \sin \theta)$$

where $a = r \cos \theta$, $b = r \sin \theta$, and $r = |z| = \sqrt{a^2 + b^2}$

In terms of Euler's formula, any complex number can therefore be written in the form

$$z = |z|e^{i\theta}$$

◇ Useful trig identities

$$\cos(-\theta) = \cos(\theta) \quad \text{and} \quad \sin(-\theta) = -\sin(\theta)$$

Example 1: Write each of the following complex numbers in standard form ($a + bi$)

(a) $e^{-i\pi/4}$

$$\begin{aligned} \text{Solution : } e^{-i\pi/4} &= e^{i(-\pi/4)} \\ &= \cos(-\pi/4) + i \sin(-\pi/4) \end{aligned}$$

(b) $e^{2e^{3+i\pi/6}}$

$$\begin{aligned} \text{Solution : } 2e^{3+i\pi/6} &= 2e^3 e^{i\pi/6} \\ &= 2e^3 [\cos(\pi/6) + i \sin(\pi/6)] \\ &= 2e^3 \cos(\pi/6) + i 2e^3 \sin(\pi/6) \end{aligned}$$

(c) e^{e^i}

$$\begin{aligned} \text{Solution : } e^{e^i} &= e^{\cos(1) + i \sin(1)} \\ &= e^{\cos(1)} e^{i \sin(1)} \\ &= e^{\cos(1)} [\cos(\sin(1)) + i \sin(\sin(1))] \\ &= e^{\cos(1)} \cos(\sin(1)) + i e^{\cos(1)} \sin(\sin(1)) \end{aligned}$$

$$(d) \frac{e^{3i} - e^{-3i}}{2i}$$

$$\begin{aligned} \text{Solution : } \frac{e^{3i} - e^{-3i}}{2i} &= \frac{(\cos(3) + i \sin(3)) - (\cos(-3) + i \sin(-3))}{2i} \\ &= \frac{[\cos(3) + i \sin(3)] - [\cos(3) - i \sin(3)]}{2i} \\ &= \frac{2i \sin(3)}{2i} \\ &= \sin(3) = \sin(3) + 0i \end{aligned}$$

Example 2: Write each of the following complex numbers in polar form ($re^{i\theta}$)

$$(a) \frac{1-i}{3}$$

Solution : Note this point would be in the 4th quadrant

$$\begin{aligned} r &= \left| \frac{1-i}{3} \right| = \left| \frac{1}{3} - \frac{i}{3} \right| = \sqrt{\frac{1}{9} + \frac{1}{9}} = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3} \\ \theta &= \tan^{-1} \left(\frac{\frac{1}{3}}{-\frac{1}{3}} \right) = \tan^{-1}(-1) = \frac{3\pi}{4} \text{ or } -\frac{\pi}{4} \end{aligned}$$

Since θ must be in the 4th quadrant $\implies \theta = -\frac{\pi}{4}$

$$\text{Therefore } \frac{1-i}{3} = \frac{\sqrt{2}}{3} e^{-i\pi/4}$$

$$(b) (1+i)^6$$

Solution : Lets convert $1+i$ first. (It is in the 1st quadrant)

$$\begin{aligned} r &= |1+i| = \sqrt{1+1} = \sqrt{2} \\ \theta &= \tan^{-1} \left(\frac{1}{1} \right) = \tan^{-1}(1) = \frac{\pi}{4} \text{ or } -\frac{\pi}{4} \end{aligned}$$

Since θ must be in the 1st quadrant $\implies \theta = \frac{\pi}{4}$

$$\text{Therefore } (1+i)^6 = \left(\sqrt{2} e^{i\pi/4} \right)^6 = (\sqrt{2})^6 e^{i6\pi/4} = 8e^{i3\pi/2}$$

$$(c) \frac{2i}{3e^{4+i}}$$

$$\text{Solution : } \frac{2i}{3e^{4+i}} = \frac{2e^{i\pi/2}}{3e^4 e^i} = \frac{2}{3e^4} e^{i\pi/2-i} = \frac{2}{3e^4} e^{i(\pi/2-1)}$$

Exercises

Using Euler's polar form only (no sines or cosines), compute the following complex powers and leave your final answer in Euler's polar form.

1. $(2 + 2i)^6$
2. $(2 + 2i)^{1/3}$
3. $4(1 - \sqrt{3}i)^7$
4. $4(1 - \sqrt{3}i)^{1/4}$
5. $\left(\frac{-2 + 2i}{3i}\right)^{1/2}$ (Hint: Write the quotient in Euler's form first)

Find all complex solutions of the following equations and leave your answers in Euler's polar form

6. $z^{1/6} - 2 - 2i = 0$
7. $z^3 + 5i = 2 + 7i$
8. $3iz^2 + 2 = 2i$